FILTER DESIGN TEST CASE

TC-SAW (Temperature Compensated SAW) - Band 2 (1900 MHz) for Millimeter-Wave Beamforming
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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1. OBJECTIVE

This document describes the design, simulation, and characterization of a TC-SAW (Temperature Compensated SAW) filter targeting Band 2 (1900 MHz) for Millimeter-Wave Beamforming. The filter is designed on 36° YX LiTaO3 substrate with a target center frequency of 4500.7 MHz and bandwidth of 105.0 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the Millimeter-Wave Beamforming module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: TC-SAW (Temperature Compensated SAW)
- Frequency Band: Band 2 (1900 MHz)
- Application: Millimeter-Wave Beamforming
- Substrate: 36° YX LiTaO3
- Package: SiP (System in Package)
- Design Tool: Sonnet EM Suite v16
- Design Center: San Jose Advanced Design Lab

This specification applies to all variants of the TC-SAW (Temperature Compensated SAW) design targeting Band 2 (1900 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency: 4500.7 MHz
Bandwidth (3 dB): 105.0 MHz
Insertion Loss Target: <= 2.7 dB
Return Loss Target: >= 19.0 dB
Out-of-Band Rejection: >= 31.0 dB
Isolation (Duplexer): >= 23.0 dB
Q Factor: 1820
Temperature Coefficient:-20.3 ppm/°C
Die Size: 2.6 x 1.3 mm
Wafer Size: 8-inch
Number of Resonators: 6
Electrode Material: Mo

Passivation: SiO2/Si3N4 stack

4. TEST PROCEDURE

1. Open Sonnet EM Suite v16 and load the TC-SAW (Temperature Compensated SAW) template project.
2. Define the substrate stack: 36° YX $LiTaO_3$ with specified layer thicknesses.
3. Set design targets: center freq 4500.7 MHz, BW 105.0 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 2.7 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (SiP (System in Package)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (8-inch, 36° YX $LiTaO_3$).
10. Perform on-wafer probing using Rohde & Schwarz ZNB40.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 13502 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 2.7 dB
- Return loss in passband shall be >= 19.0 dB
- Out-of-band rejection at +/- 158 MHz offset >= 31.0 dB
- Group delay variation across passband shall be <= 8 ns
- Temperature drift over -40°C to +85°C shall be <= 11.4 MHz
- Power handling shall be >= +25 dBm at 1 dB compression
- ESD tolerance >= 1000 V (HBM)
- Die yield on qualification lot >= 92%

6. TEST RESULTS

Measurement Date: 2024-05-04
Devices Tested: 89
Insertion Loss (meas): 2.39 dB
Return Loss (meas): 20.9 dB
Rejection (meas): 33.6 dB
Isolation (meas): 28.4 dB
Q Factor (meas): 1820
Group Delay Variation: 2.3 ns
Power Handling: +30 dBm
Die Yield: 91.8%

Result Classification: FAIL

7. OBSERVATIONS AND FINDINGS

a) The TC-SAW (Temperature Compensated SAW) design demonstrated below-target performance for Band 2 (1900 MHz).
b) Insertion loss met target specification.
c) Out-of-band rejection exceeded the minimum requirement.
d) Temperature stability was within specification across full range.
e) Wafer-level uniformity was excellent (sigma < 1.5%).

8. CORRECTIVE ACTIONS

1. Optimize resonator geometry to reduce insertion loss by ~0.3 dB.
2. Re-run EM simulation with refined mesh for better accuracy.
3. Escalate to advanced materials team for alternative piezoelectric stack.

9. SIGN-OFF

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